Geometric Bogomolov Conjecture for Semiabelian Varieties

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- Let $M := \mathcal{O}_{\overline{G}}(\overline{G} \setminus G)$, and N be a symmetric ample line bundle on A. Then the ample line bundle $L := M + \overline{\pi}^* N$ gives a **canonical height** $\widehat{h}_L : G(\overline{K}) \to \mathbb{R}_{\geq 0}$.

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- **Goal:** Classify subvarieties of $G_{\overline{K}}$ with Zariski dense small points.

Main Theorem

Theorem (Geometric Bogomolov Conjecture for semiabelian varieties)

For a closed subvariety $X \subset G_{\overline{K}}$, the following are equivalent:

(a) *X contains Zariski dense sets of small points w.r.t. $\widehat{h}_L$.*

(b) *X is special: after quotienting out $\mathrm{Stab}(X)$, it becomes a torsion translate of a constant variety defined over k.*

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- This result is independent of the choice of compactification, since the heights induced by different compactifications are comparable.
- This generalized the abelian case. And we have a new phenomenon only shows up in our case:
  X is special$\Leftrightarrow$ small points are dense $\not\Rightarrow$ points of height 0 are dense.

A look-back for proofs of GBC for abelian varieties

- Using the Poincaré reducibility, **(GBC)** for abelian varieties was proved part by part in the arbitrary characteristic case.
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- In the characteristic 0 case, **(GBC)** was proved by using Betti-maps as in Gao–Habegger, Cantat–Gao–Habegger–Xie.

Obstacles in the semiabelian case

- **No Poincaré reducibility** for semiabelian varieties:
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  - **This fails over function fields!**
- In the characteristic 0 case, there were no Betti-maps. (We recently defined Betti-maps for families of semiabelian varieties and generalized the uniform Mordell-Lang conjecture to semiabelian varieties in a joint work with Zhaobo Han.)

Outline of the Proof

1. **Quasi-split case:** G isogenous to $G_0 \times_k A_1$, with A_1 of trivial trace.
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   - Generalize Yamaki's theory of *relative heights*.
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2. **General case:** reduction to the quasi-split case via *relative Faltings–Zhang maps*:

$$\begin{array}{ccc} G^n_{/A} & \xrightarrow{\alpha_n} & \mathbb{G}_m^{t(n-1)} \times A \\ & \searrow^{\beta_n} & \uparrow{\scriptstyle \mathrm{id}\times\pi} \\ & & \mathbb{G}_m^{t(n-1)} \times G \end{array}$$

where $\alpha_n(x_1, \ldots, x_n) := (x_1 - x_2, \ldots, x_{n-1} - x_n, \pi(x_1))$ and $\beta_n(x_1, \ldots, x_n) := (x_1 - x_2, \ldots, x_{n-1} - x_n, x_n)$.

Key ingredients of the relative Faltings-Zhang reduction

- **A modification of Zhang's fundamental inequality** works for both function fields and number fields

$$\mathrm{ess}_{\overline{L}}(X) \geq h_{\overline{L}}(X) \geq \frac{1}{\dim X + 1}(\mathrm{ess}_{\overline{L}}(X) + \dim X \cdot \underbrace{\lambda_{\min}(\overline{L})}_{\text{new invariant}})$$

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Thank you!